

CIROH Hydrologic Prediction Testbed

Book of Protocols

October 2024



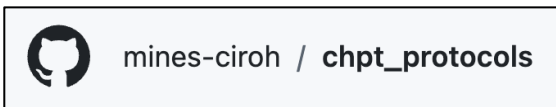
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Overview

The US NOAA Cooperative Institute for Research to Operations in Hydrology (CIROH) is a consortium of US institutions that partner with the US National Water Center (NWC) to conduct research to advance operational hydrologic forecasting services. One CIROH project at the Colorado School of Mines (CSM) is the CIROH Hydrologic Prediction Testbed (CHPT) – a community-oriented initiative to establish rigorous, quantitative intercomparison and benchmarking of US operational hydrologic forecasts, and particularly the multiple elements – models, methods, datasets – involved in producing them. The Testbed’s overarching goal is to address the fragmentation of research into fundamental challenges and needs for operational prediction systems, which is a significant impediment to integrated review and decision-making across diverse investments into forecasting R&D.

The CHPT comprises multiple community-developed experimental **protocols** with standardized experiments, benchmarks and evaluation, targeting different forecasting and forecasting sub-component objectives. Examples of sub-components include model formulations, model parameterizations (e.g., a glacier physics algorithm), input datasets, and techniques (e.g., data assimilation, post-processing, ensemble methods), across timescales from nowcasting to multi-season prediction.

This booklet contains examples of several different protocols being proposed for the CHPT, which will also be available in machine-readable formats at the public GitHub repository below. The effort is open to collaborators both within and beyond CIROH. ***The protocols are neither final nor complete and are presented here to invite feedback, discussion, and engagement.***



This project was supported by the NOAA Cooperative Institute for Research to Operations in Hydrology (CIROH), which is funded through the NOAA Cooperative Agreement with The University of Alabama (NA22NWS4320003). The CIROH subproject supporting the CHPT is entitled "CIROH Hydrometeorological Prediction Testbed", awarded in 2022 and led by Andy Wood at the Colorado School of Mines. This booklet was prepared by Josh Sturtevant and Andy Wood, with contributions from Ethan Ritchie.

1.1. Testbed linkages with other CIROH components and projects

The CHPT depends on linkages with other components or aspects of CIROH, and also leverages external resources, including datasets and expertise from agencies and groups related to operational Earth System monitoring and prediction. Within CIROH, the Testbed project coordinates with three key activities (**Figure 1-1**):

- **Model and Forecast Evaluation.** CHPT will co-design forecast-related evaluation experiments and workflows to make use of the RTI evaluation tools (namely TEEHR; <https://github.com/RTIInternational/teehr>), as appropriate.
- **CIROH Cyber-Infrastructure (C-I).** CHPT will try to steer and/or facilitate the use of the C-I to enable CIROH teams to test innovations and generate outputs matching the CHPT protocols. The C-I and data science theme projects will provide data services and computational resources that can be used in forecast evaluations.
- **CIROH Project Advances.** The ultimate value and usefulness of the CHPT depends on the engagement of CIROH project teams to help develop and design the protocols and assess new innovations in the associated benchmark-oriented experiments.

The CHPT will also coordinate with other efforts and groups, including (but not limited to) OWP and NWC, NWS River Forecast Centers, WMO, CIROH leadership, and CIROH RT4 (stakeholder communication and product activities), which will help to translate quantitative CHPT insights into messaging that resonates with forecast users.

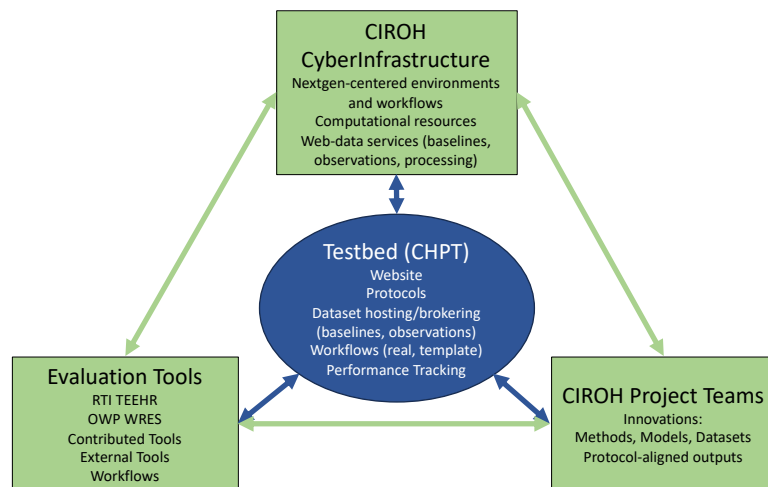


Figure 1-1: Linkages of CIROH Hydrologic Prediction Testbed (CHPT) with other elements of CIROH.

1.2. Engaging with the CHPT

CIROH project teams, external collaborators and interested parties can participate in the CHPT through identifying an existing experimental protocol (or proposing a new community protocol) that aligns with their own research questions or activities. The protocol lead(s) or core CHPT team can be contacted to discuss project activities and assess the feasibility of running experiments supporting the protocol, as well as potential refinements to the protocol to support inclusion. Note, as protocols become established, they become more difficult to change without affecting the consistency of prior results.

CHPT Governance and Activities

The CHPT is part of CIROH Project 1 (at the Colorado School of Mines) and is a core component of CIROH Theme 1 – Advancing Operational Prediction. A group of leads from each of the linked activities and CIROH RTs will meet semi-annually to review CHPT progress, and open webinar/discussion meetings will be held to highlight CHPT activities and engage the community (CIROH and external). The CIROH Slack testbed channel is also available to CIROH partners to connect with CHPT collaborators.

2. Protocols

2.1. Protocol Elements

The CHPT is made up of a collection of standardized experimental protocols for evaluating and benchmarking **innovations** in different forecast related capabilities. Each CHPT protocol contains a number of universal elements (listed below). Templates for these elements, as well as for the individual innovations or experiments being evaluated within a given protocol, are included in Section 3.

Objective or focus: e.g., predictands, analyses, forecast information

Observational datasets that can be used for validation and verification

Reference capabilities against which to assess strength of innovation (i.e., the baselines)

Experimental design: the periods, catchments or domain, lead times if applicable, and other relevant details to ensure consistency in evaluating candidate innovations

Metrics of performance (absolute and relative)

Other factors to be taken into consideration in the capability evaluation (e.g., computational expense, portability, complexity, dependencies, CONUS-wide relevance, potential operational latency, and so on)

Key references, studies, and relevant external activities

Contact(s): Lead(s) or active contributors, e.g., helping connect new participants, identify new relevant work, ensuring that testbed results are up-to-date, and so on.

Metadata: Machine-readable information structure to exactly identify each element used in each innovation within a protocol, including details such as model or dataset version, algorithm variation, parameter set variation, generation date, author, among others.

P-1: Streamflow simulation in headwater catchments

Status: ready, targeting protocol group formation in 2025

Protocol Element	Description
Focus / Objective	Retrospective streamflow simulation performance across multiple verification features
Observation(s)	Gage measurements of flow at hourly/daily timesteps
Reference capabilities (baselines)	- NWS RFC SAC-SMA/Snow17 - NWM 3.0 streamflow forecasts - Nextgen-based NWM default configuration
Experimental design	- Spatial domain: CAMELS catchments and/or various subsets (e.g., 516 hourly gages, RFC-forecasted) - Validation Period: 10 years of daily streamflow simulations, 1990-1999 - Timestep(s): daily, 6-hourly, hourly - Types of validation: (1) temporal across all basins; (2) spatial cross-validation (CV); (3) spatial & temporal CV
Metrics of performance	- Peak flow magnitude and timing; cumulative flow vol. - Modified KGE, NSE - ETS for flow versus defined thresholds (e.g., flood) - Accuracy metrics: correlation, rBias, RMAE 4-quadrant error charts - Metrics also expressed relative to the reference capability
Other considerations	- Computational demand of innovation - Stability of dependencies (method, dataset) - Potential operational latency - Extent of documentation and published research - Potential for forecaster training and real-time support - Dependency on proprietary resources or methods
Metadata	- Institutional association and project lead - Versioning, date of generation, parameter set and/or algorithm variation, etc.
Contacts	J. Sturtevant, A. Wood, C. Shen, J. Frame, M. Clark, K. Van Werkhoven

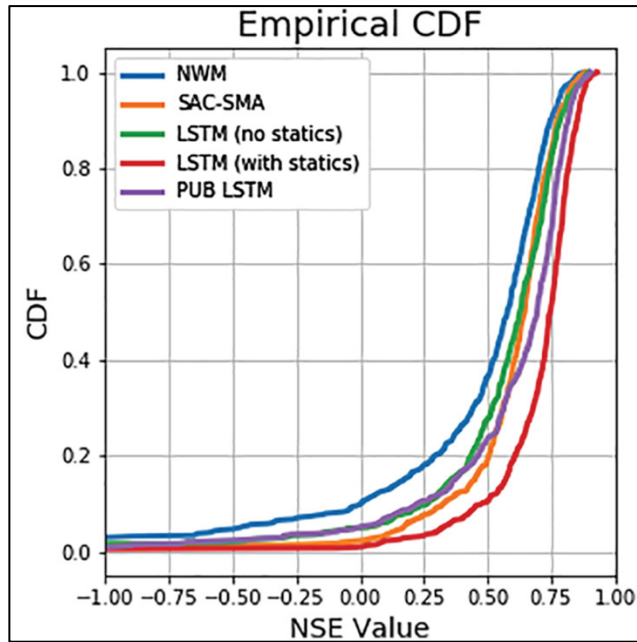


Figure 2-1: Example streamflow prediction benchmarking study comparing the cumulative distribution function of NSE skill scores from various retrospective streamflow simulation methods across the CAMELS basins (Kratzert et al., 2019).

P-2: Short-range (1-7 day) deterministic flow forecast in headwater catchments

Status: ready, targeting protocol group formation in 2025

Protocol Element	Description
Focus / Objective	Short-range (1-7 day) deterministic flow forecast performance across multiple verification features
Observation(s)	Gage measurements of flow at hourly/daily timesteps
Reference capabilities (baselines)	- RFC streamflow forecasts - NWM 3.0 streamflow forecasts - Nextgen-based NWM short-range configuration - Persistence forecast
Experimental design	- Spatial: Subset of CAMELS catchments that is forecast by RFC combined with - Study Period: 5 years of daily hindcasts from approximately 6Z (RFC init.), 2017-2022 - Lead Times: 1-7 days - Timestep(s): daily, 6-hourly, hourly
Metrics of performance	- Peak flow magnitude and timing - Modified KGE of forecast flow for each lead time - Cumulative flow volume at each daily lead time - ETS for flow versus defined thresholds (e.g., flood) - Bias, correlation, MAE for each lead time - Metrics also expressed relative to reference capability
Other considerations	- Computational demand of innovation - Stability of dependencies (method, dataset) - Potential operational latency - Extent of existing documentation, published research - Potential for forecaster training and real-time support - Dependency on proprietary resources or methods
Metadata	- Institutional association and project lead - Versioning, date of generation - Parameter set and/or algorithm variation, etc.
Contacts	J. Sturtevant, T. Dixon, A. Wood, ...

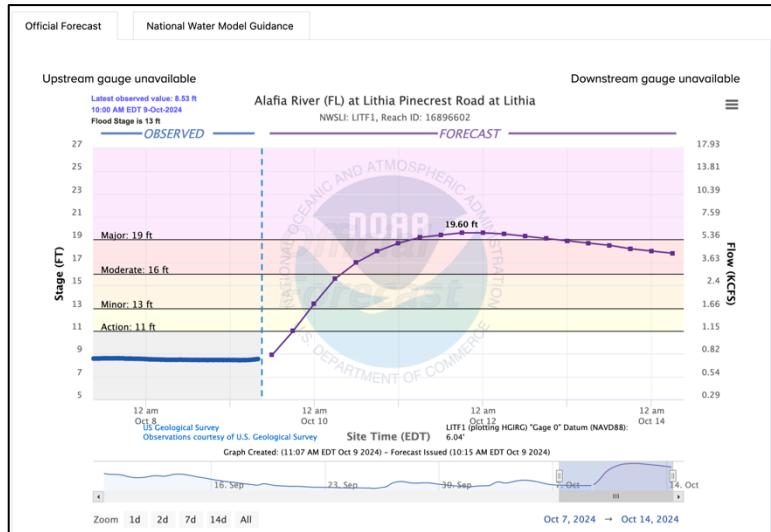


Figure 2-2: Example short-range streamflow forecast from NOAA's Southeast River Forecast Center (SERFC). NOAA-RFC forecasts will serve as a reference capability against which to benchmark new innovations.

P-3: Catchment areal average daily snow water equivalent (SWE) analysis

Status: ready, targeting protocol group formation in 2025

Protocol Element	Description
Focus / Objective	Retrospective SWE simulation performance verified by ASO observations in the hydrofabric
Observation(s)	• SNOTEL stations • Airborne Snow Observatory (ASO) lidar flights
Reference capabilities (baselines)	• NOAA NOHRSC • SNODAS
Experimental design	• Spatial domain: ASO basin extents largely in CA & CO • Study Period: 12 years of irregular ASO LiDAR flights, 2013-2024 • Lead Times: retrospective • Timestep(s): Irregular, dates of ASO LiDAR flights.
Metrics of performance	• Cumulative SWE volume • Mean difference of cumulative SWE • KGE • RMSE of peak SWE for scenes. • R-squared of modeled vs. observed SWE • Metrics also expressed relative to reference capability
Other considerations	• Computational demand of innovation • Stability of dependencies (method, dataset) • Potential operational latency • Extent of existing documentation, published research • Potential for forecaster training and real-time support • Dependency on proprietary resources or methods
Metadata	• Institutional association and project lead • Versioning, date of generation • Parameter set and/or algorithm variation, etc.
Contact(s)	• Andy Wood, Ethan Ritchie, Ryan Johnson

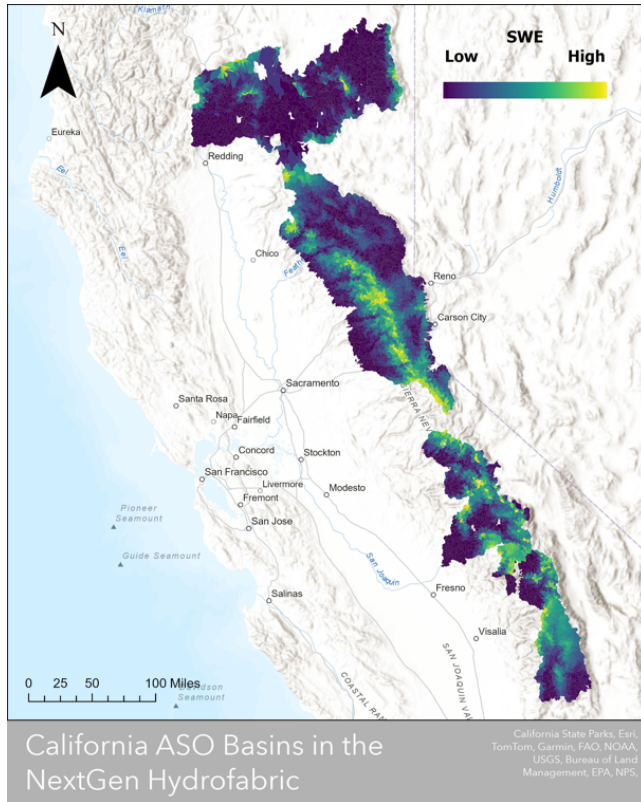


Figure 2-3: Example of ASO-based observations over the NextGen hydrofabric in the California Sierra Mountains (data obtained from ASO, Inc. and processed at Mines).

P-4: Seasonal basin-scale precipitation and temperature

Status: early stages/conceptual

Protocol Element	Description
Focus / Objective	Seasonal (month 2-11 and seasons 1-10) climate forecast performance at a sub-regional basin-scale (HUC4) for temperature (T) and precipitation (P)
Observation(s)	Gridded reanalysis datasets including AORC, NLDAS, and ERA5-Land
Reference capabilities (baselines)	- NOAA Climate Prediction Center (CPC) - NMME hindcasts (especially models from US centers, e.g., NCEP-CFSv2)
Experimental design	- Spatial domain: CONUS sub-regional (HUC4) basins - Study Period: 28 years of monthly hindcasts (1982-2010) - Lead Times: months 2-11, seasons 1-10 - Timestep(s): monthly, seasonal
Metrics of performance	- CPRSS (probabilistic) - ACC (deterministic) - All metrics are also expressed as skill scores relative to the reference capability
Other considerations	- Computational demand of innovation - Stability of dependencies (method, dataset) - Potential operational latency - Extent of documentation and published research - Potential for forecaster training and real-time support - Dependency on proprietary resources or methods
Metadata	- Institutional association and project lead - Versioning, date of generation - Parameter set and/or algorithm variation, etc.
Contact(s)	J. Sturtevant, A. Wood, ...

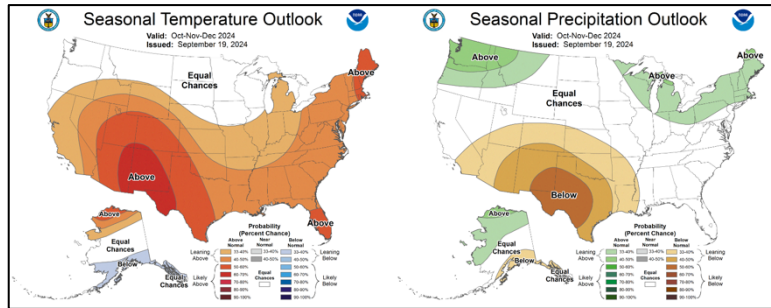


Figure 2-4: Example NOAA Climate Prediction Center (CPC) seasonal outlooks for temperature (left) and precipitation (right). NOAA CPC seasonal climate forecasts will serve as a reference capability against which to benchmark new innovations.

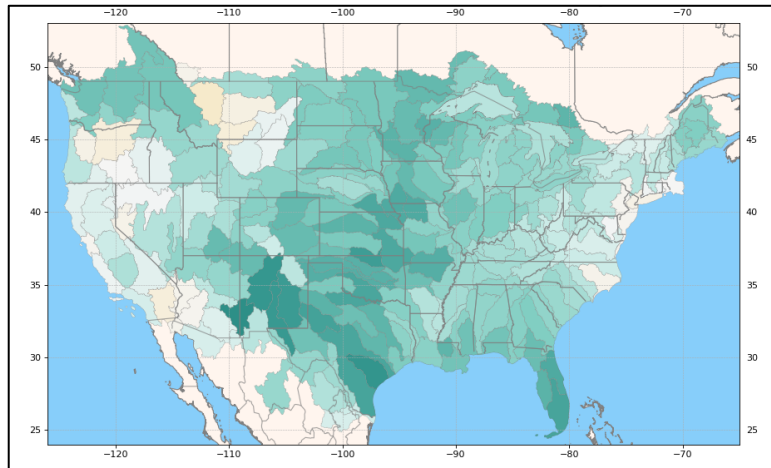


Figure 2-5: Example of the USGS HUC4-based S2S climate testbed analysis configuration, which is being adapted from an existing S2S climate testbed that has been created at NCAR (sponsored by USACE, Reclamation, and NOAA MAPP).

P-5: Seasonal water supply forecasting (volumetric and daily hydrographs)

Status: early stages/conceptual, needs protocol lead

Protocol Element	Description
Focus / Objective	Seasonal streamflow forecast performance across multiple verification features
Observation(s)	Point gage measurements of April-July flow
Reference capabilities (baselines)	- NOAA-RFC water forecasts (daily hydrograph) - USDA-NRCS water supply forecasts (volumetric only) - Climatology
Experimental design	- Spatial: Subset of CAMELS catchment that is forecast by western RFCs - Study Period: 30 years of April 1st hindcasts, 1991-2020 - Lead Times: 1-12 months - Predictand: daily, seasonal, common WSF periods (e.g., Apr-July, Apr-Sep)
Metrics of performance	- MAE, RMSE, percent bias of AMJJ volumes - Reliability diagram and contingency-based metrics - KGE/NSE for ESP-type methods with daily timesteps - Metrics also expressed relative to reference capability
Other considerations	- Computational demand of innovation - Stability of dependencies (method, dataset) - Potential operational latency - Extent of existing documentation and published research - Potential for forecaster training and real-time support - Dependency on proprietary resources or methods
Metadata	- Institutional association and project lead - Versioning, date of generation - Parameter set and/or algorithm variation, etc.
Contacts	TBD

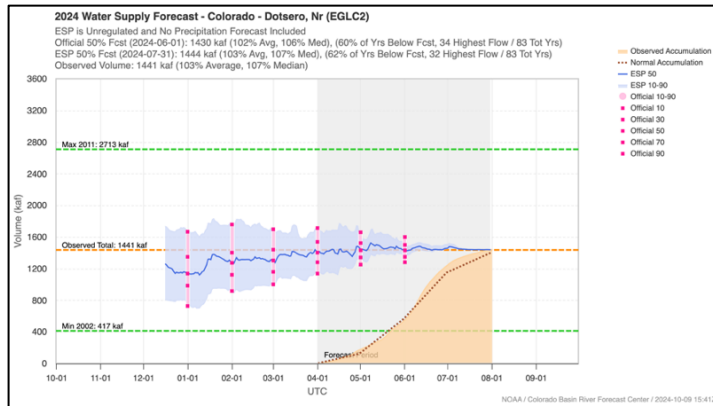


Figure 2-6: Example NOAA-CBRFC water supply forecast for the Colorado River at Dotsero, CO. NOAA-RFC forecasts will serve as a reference capability against which to benchmark new innovations.

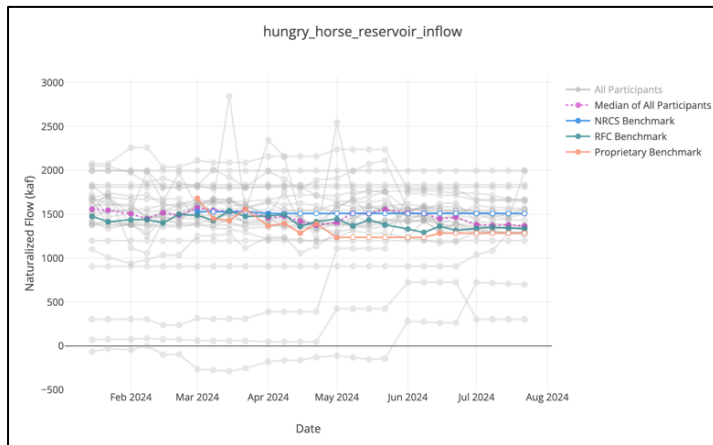


Figure 2-7: Example water supply forecast benchmarking from the 2024 Bureau of Reclamation Water Supply Forecast Rodeo, which awarded over \$400,000 in prizes for the best forecasts.

P-6: Climate prediction: sub-seasonal basin-scale precipitation and temperature

Status: early stages/conceptual, needs protocol lead

Protocol Element	Description
Focus / Objective	Sub-seasonal (days 8-45) climate forecast performance at a sub-regional basin-scale (HUC4) for temperature (T) and precipitation (P)
Observation(s)	Gridded reanalysis datasets including AORC, NLDAS, and ERA5-Land
Reference capabilities (baselines)	- NOAA Climate Prediction Center (CPC) - GEFS hindcasts
Experimental design	- Spatial domain: CONUS sub-regional (HUC4) basins - Study Period: 26 years of weekly hindcasts (1991-2017) - Lead Times: 1, 2, 3 weeks - Timestep(s): weeks 2-3, 3-4, 5-6, 2-4.
Metrics of performance	- CPRSS (probabilistic) - ACC (deterministic) - All metrics are also expressed as skill scores relative to the reference capability
Other considerations	- Computational demand of innovation - Stability of dependencies (method, dataset) - Potential operational latency - Extent of documentation and published research - Potential for forecaster training and real-time support - Dependency on proprietary resources or methods
Metadata	- Institutional association and project lead - Versioning, date of generation - Parameter set and/or algorithm variation, etc.
Contact(s)	TBD

P-7: Point-based snow water equivalent (SWE) analysis

Status: early stages/conceptual, needs protocol lead

Protocol Element	Description
Focus / Objective	Retrospective SWE simulation performance verified by SNOTEL point-based observations
Observation(s)	• SNOTEL point measurements of SWE at daily timesteps
Reference capabilities (baselines)	• NOAA NOHRSC • SNODAS
Experimental design	• Spatial domain: Western US SNOTEL stations • Validation period: 12 years of daily SWE measurements (2001-2010) • Lead Times: retrospective • Timestep(s): daily
Metrics of performance	• Peak SWE magnitude and timing • KGE • Cumulative SWE volume • SWE melt-out timing • All metrics are also expressed as skill scores relative to the reference capability
Other considerations	• Computational demand of innovation • Stability of dependencies (method, dataset) • Potential operational latency • Extent of existing documentation, published research • Potential for forecaster training and real-time support • Dependency on proprietary resources or methods
Metadata	• Institutional association and project lead • Versioning, date of generation • Parameter set and/or algorithm variation, etc.
Contact(s)	• TBD

P-8: Gridded daily surface meteorological forcings

Status: early stages/conceptual, needs protocol lead

Protocol Element	Description
Focus / Objective	• Evaluate the quality of precipitation and temperature forcings for streamflow and SWE simulation using a trained ML model
Observation(s)	• CAMELS gage observed daily/hourly flow • SNOTEL point SWE observations
Reference capabilities (baselines)	• AORC
Experimental design	• Train and test two LSTM models, one for streamflow and SWE, using different alternative forcings • Testing based on a 5-fold spatial cross-validation and temporal crossvalidation • Period 2001-2010
Metrics of performance	• Modified KGE • Accuracy metrics: bias, correlation, RMAE • Conditional scores (different quantiles) • All metrics are also expressed as skill scores relative to the reference capability
Other considerations	• Latency and reliability of forcing dataset • Retrospective period of dataset • Consistency of real-time and retrospective data • Accessibility and quality of dataset production methods
Metadata	• Institutional association and project lead • Version info, date of generation • Parameter set and/or algorithm variation, etc.
Contact(s)	• Juliette Mukangango, A. Wood, ...

3. Templates

The CIROH Hydrologic Prediction Testbed (CHPT) aims to include protocols that span the range of hydrologic prediction elements. Many other protocols than the examples listed here are possible. Ideally, as has occurred organically around the first protocol, a community converges to adopt a shared, common experiment and standards through which their research outcomes can be evaluated and benchmarked.

Below are templates that can be used to propose/establish new protocols and document the innovations that would be assessed.

Have an idea for a CHPT protocol you would like to organize? Write in ideas to share with the project team below!

3.1.1. Protocol Template

Table 3-1: Protocol template

Protocol Element	Description
Focus / Objective	
Observation(s)	
Reference capabilities (baselines)	
Experimental design	
Metrics of performance	
Other considerations	
Metadata	
Contact(s)	

CHPT contacts at the Colorado School of Mines

- Josh Sturtevant, joshua.sturtevant@mines.edu
- Andy Wood, awwood@mines.edu
- Terri Hogue, thogue@mines.edu
- Ethan Ritchie, eritchie@mines.edu

Key Collaborators

- Katie Van Werkhoven and Matthew Denno, RTI
- James Halgren, U. of Alabama
- Projects: C. Shen, PSU; R. Johnson, UU; M. Clark (U. Calgary)